

CSA1712

ARTIFICIAL INTELLIGENCE

NAME: VARSHINI M

REG NO.: 192511215

Question 1 – Rain and Wet Ground

Aim

To prove that the ground is wet using the **Resolution Algorithm**.

Knowledge Base

1. If it rains, then the ground becomes wet.
2. It is raining.

Goal

Prove that the ground is wet.

Step 1: Propositional Representation

Let

- R = It is raining
- W = Ground is wet

Knowledge Base:

1. $R \rightarrow W$
2. R

Goal: W

Step 2: Convert into CNF

Using $P \rightarrow Q \equiv \neg P \vee Q$

- $C1 : \neg R \vee W$
- $C2 : R$

Step 3: Negate the Goal

Goal = W

Negated Goal:

- $C3 : \neg W$

Step 4: Resolution

Resolve C1 and C2:

$(\neg R \vee W), (R)$

$\Rightarrow C4 : W$

Resolve C4 and C3:

$(W), (\neg W)$

$\Rightarrow \square$ (Empty Clause)

Conclusion

Since the **Empty Clause** ($\square$) is obtained, the negation of the goal is false.

Therefore, the ground is wet is proved.

Question 2 – Student Assignment Submission

Aim

To prove that Rahul receives internal marks using Resolution.

Knowledge Base

1. If a student submits the assignment, then the student receives internal marks.
2. Rahul submitted the assignment.

Goal

Prove that Rahul receives internal marks.

Step 1: Propositional Representation

- S = Rahul submitted assignment
- M = Rahul receives internal marks

Knowledge Base:

1. $S \rightarrow M$
2. S

Goal: M

Step 2: CNF

- $C1 : \neg S \vee M$
- $C2 : S$

Step 3: Negate Goal

- $C3 : \neg M$

Step 4: Resolution

Resolve C1 and C2:

$(\neg S \vee M), (S)$

$\Rightarrow C4 : M$

Resolve C4 and C3:

$(M), (\neg M)$

$\Rightarrow \square$

Conclusion

Empty Clause is obtained.

Hence, Rahul receives internal marks is proved.

Question 3 – Library Membership

Aim

To prove that Priya can borrow books using Resolution.

Knowledge Base

1. If a person is a library member, then the person can borrow books.
2. Priya is a library member.

Goal

Prove that Priya can borrow books.

Step 1: Propositional Representation

- L = Priya is a library member
- B = Priya can borrow books

Knowledge Base:

1. $L \rightarrow B$
2. L

Goal: B

Step 2: CNF

- $C1 : \neg L \vee B$
- $C2 : L$

Step 3: Negate Goal

- $C3 : \neg B$

Step 4: Resolution

Resolve C1 and C2:

$(\neg L \vee B), (L)$

$\Rightarrow C4 : B$

Resolve C4 and C3:

$(B), (\neg B)$

$\Rightarrow \square$

No further clauses can be generated.

Conclusion

Empty Clause is derived.

Therefore, Priya can borrow books is proved.

Question 4 – Placement Eligibility

Aim

To prove that Arun is eligible for placement using Resolution.

Knowledge Base

1. If a student clears the aptitude test, then the student is eligible for placement.
2. Arun cleared the aptitude test.

Goal

Prove that Arun is eligible for placement.

Step 1: Propositional Representation

- A = Arun cleared aptitude test
- P = Arun is eligible for placement

Knowledge Base:

1. $A \rightarrow P$

2. A

Goal: P

Step 2: CNF

- $C1 : \neg A \vee P$
- $C2 : A$

Step 3: Negate Goal

- $C3 : \neg P$

Step 4: Resolution

Resolve C1 and C2:

$(\neg A \vee P), (A)$

$\Rightarrow C4 : P$

Resolve C4 and C3:

$(P), (\neg P)$

$\Rightarrow \square$

Resolution Tree

$\neg A \vee P$
|
| Resolve with A
|
P
|
| Resolve with $\neg P$
|
 $\square$

Conclusion

Empty Clause is obtained.

Hence, Arun is eligible for placement is proved.

Question 5 – Access Control System

Aim

To prove that the user is granted access using Resolution.

Knowledge Base

1. If a user enters the correct password, then the user is authenticated.
2. If a user is authenticated, then the user is granted access.
3. The user entered the correct password.

Goal

Prove that the user is granted access.

Step 1: Propositional Representation

- P = Correct password entered
- A = User authenticated
- G = User granted access

Knowledge Base:

1. $P \rightarrow A$
2. $A \rightarrow G$
3. P

Goal: G

Step 2: CNF

- $C1 : \neg P \vee A$
- $C2 : \neg A \vee G$
- $C3 : P$

Step 3: Negate Goal

- $C4 : \neg G$

Step 4: Resolution

Resolution 1

$(\neg P \vee A), (P)$

$\Rightarrow C5 : A$

Resolution 2

$(\neg A \vee G), (A)$

$\Rightarrow C6 : G$

Resolution 3

$(G), (\neg G)$

$\Rightarrow \square$

Conclusion

Empty Clause is obtained.

Therefore, the user is granted access is proved using the Resolution Algorithm.

Overall Result

All the given goals are successfully proved using the **Resolution Algorithm**, and in each case the derivation of the **Empty Clause ($\square$)** confirms that the goal logically follows from the Knowledge Base. This demonstrates the use of **resolution-based automated reasoning in Artificial Intelligence**.